

Alkenes

Alkenes are **unsaturated hydrocarbons** with at least one double bond hence they do **not** contain the maximum amount of hydrogen.

General formula: C_nH_{2n}

Properties

- 1-**Insoluble** in water due to their non-polar nature, but soluble in non-polar organic solvents
- 2-**Flammable** and burn with a smoky flame due to higher carbon content relative to alkanes.
- 3-**More reactive than alkanes**, undergoing addition reactions (e.g. with bromine, hydrogen, steam) due to the presence of the carbon-carbon double bond ($C=C$).
- 4-*States (C_2 to C_4 : Gases / C_5 to C_{18} : Liquids / C_{19} and above: Solids) at room temperature.
- 5-*Boiling/Melting Points increase with chain length due to increased van der Waals forces, but are slightly lower than equivalent alkanes due to reduced molecular symmetry.
- 6-Viscosity increases with chain length due to stronger intermolecular forces.
- 7-Typically **colourless** and can have faint, sweetish or petrol-like odours.

Use

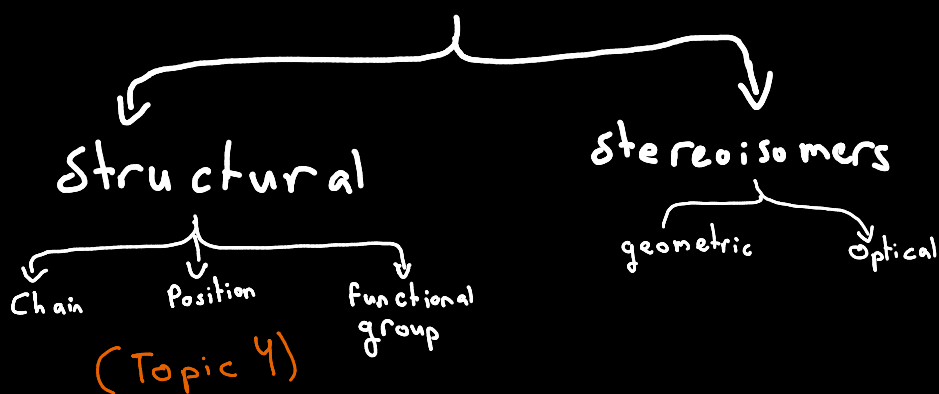
- 1-Making polymers
- 2-ripening / maturing fruit e.g. bananas (**ethene specific** use)

The double bond

A double bond is not two identical bonds. It is a combination of two different types of covalent bonds: one σ bond and one π bond (**explained in more details in chapter 3**).

Feature	Sigma (σ) Bond	Pi (π) Bond
Formation	Head-on overlap of orbitals	Sideways overlap of p-orbitals
Strength	Strong	Weaker
Rotation	Allows free rotation	Locks atoms, prevents rotation
Electron Location	Directly between nuclei	Above & below the bond axis
Reactivity	Very stable, unreactive under normal conditions	Electron-rich , vulnerable to attack by electrophiles

Types of isomers

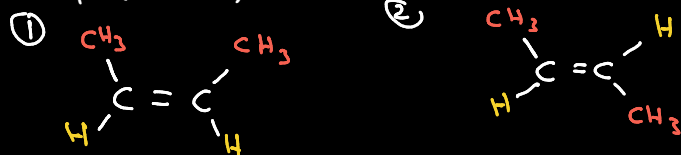


Stereoisomers: compounds same molecular and structural formulae but atoms or groups arranged **differently in 3D**.

Geometric Isomers

- Difference: compound containing a $C=C$ with atoms or groups attached at different positions.
- Cause:
- 1- **Restricted rotation** about the carbon-carbon **double bond** (π bond).
 - 2- There are two **different groups** on **each of the carbon atoms** in the $C=C$ bond / double bond

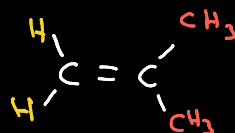
Example: (But-2-ene)



When Alkenes CANNOT Have Geometric Isomers

Condition: An alkene cannot have geometric isomers if at least one of the double-bonded carbons has **two identical substituents** (groups or atoms).

Example:



Note: Alkanes can't have stereoisomers as they only have single bonds

Cis and Trans naming system

- Look for two identical groups or atoms, one attached to each carbon of the double bond.
- Cis (means "same side"): The two identical groups are on the **SAME SIDE** of the molecule.
 - Trans (means "across"): The two identical groups are on **OPPOSITE SIDES**.

Cis and Trans naming **ONLY** works when both of the following are true:

- Each carbon of the double bond has two different groups attached (so rotation matters).
- There are **two identical** substituents, one on each carbon of the double bond, that can be directly compared.

If this condition is not met, the system fails.

E/Z naming system

Unlike cis/trans (which compares identical groups), E/Z uses a priority system. It compares the higher-priority group on each carbon of the double bond.

Process:

- For **EACH** carbon of the double bond, independently assign priority to its two attached groups using the Cahn-Ingold-Prelog (CIP) Rules.
- Identify the higher-priority group on Carbon 1 and the higher-priority group on Carbon 2.

Look at the spatial arrangement of these two high-priority groups:

Z (from German zusammen, meaning "together"): The two high-priority groups are on the **SAME SIDE**.

E (from German entgegen, meaning "opposite"): The two high-priority groups are on **OPPOSITE SIDES**.

The Priority Rules

Rule 1: Atomic Number

Look at the atom directly attached to the double bond carbon.

Higher **atomic number** = higher priority.

Example: Br (35) > Cl (17) > S (16) > F (9) > O (8) > N (7) > C (6) > H (1)

Rule 2: If First Atoms Are Identical (Tie)

Move outward to the next set of atoms connected to those first atoms.

Compare their atomic numbers as a list, in descending order.

Example: $-CH_3$ vs $-CH_2CH_3$

First atoms: Both are C (tie).

Move out:

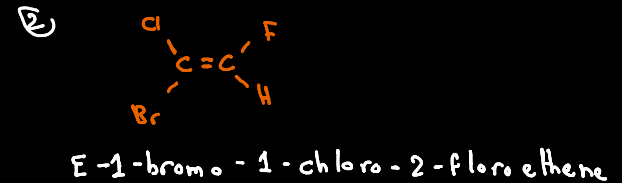
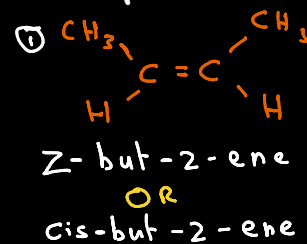
$-CH_3$ attaches to (H, H, H) $\rightarrow$ atomic numbers (1, 1, 1)

$-CH_2CH_3$ attaches to (C, H, H) $\rightarrow$ atomic numbers (6, 1, 1)

Compare: (6,1,1) vs (1,1,1) $\rightarrow 6 > 1$

Result: $-CH_2CH_3$ has higher priority than $-CH_3$.

Examples



③



E-pent-2-ene

Reactions

Introduction

Alkenes are a homologous series of **unsaturated hydrocarbons** characterized by having at least one carbon-carbon double bond (C=C). The presence of this π (pi) bond, which is **electron-rich and relatively weak, makes alkenes much more reactive than alkanes**. They primarily undergo **electrophilic addition reactions**, where the π bond breaks open and two new atoms or groups add to the carbons.

Addition reaction: reaction in which two molecules combine to form one molecule

Electrophile: chemical species that is electron-deficient and seeks to accept a pair of electrons to form a new bond (electron lover)

Types of electrophiles:

- 1) Positively charged ions (cations like H^+ , NO_2^+)
- 2) Molecules with a partial positive charge (δ^+ , like the H in H-Br)
- 3) Molecules with an incomplete octet, e.g. (BF_3 , carbocations)

Electrophiles with alkenes

Alkenes have an **electron-rich π bond** (a cloud of electrons above and below the C=C). **Electrophiles** are drawn to and attack this electron-rich area, starting the reaction.

Example: In the reaction of ethene with HBr:

HBr is **polar**: H is δ^+ , Br is δ^- .

The δ^+ H acts as the **electrophile**: It is **attracted to the π electrons** of the double bond.

The H accepts the electron pair from the π bond, breaking the π bond and forming a new C-H bond.

Addition of Hydrogen (Hydrogenation)

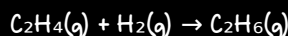
–Reagents & Conditions: Hydrogen gas (H_2), **Nickel** (Ni) catalyst.

–Mechanism Type: **addition/reduction** (gaining hydrogen reduces the alkene).

–What Happens: The H-H bond breaks, and a hydrogen atom adds to each carbon of the double bond.

–Product: An alkane (saturated hydrocarbon).

–Equation Example:



Ethene $\rightarrow$ Ethane

Use: To harden vegetable oils (unsaturated fats) into margarine (saturated/partially saturated fats).
note: vegetable oils are partially hydrogenated so not all double bonds are broken

Addition of Halogens (Halogenation)

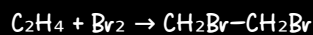
–Reagents: Bromine (Br_2) or Chlorine (Cl_2)

–Mechanism: Electrophilic addition.

–What Happens: The halogen molecule (X-X) is polarized and adds across the double bond.

–Product: A di-substituted halogenoalkane (dihaloalkane).

Equation Example:



Ethene $\rightarrow$ 1,2-Dibromoethane (colorless)

Observable Test: Decolorization of **brown** liquid bromine (or **orange** aqueous bromine). This is the standard test for unsaturation (C=C or $\text{C}\equiv\text{C}$ bonds).
if added to alkanes the color will remain brown

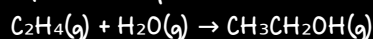
Addition of Steam (Hydration)

–Reagents & Conditions: Steam ($\text{H}_2\text{O}(\text{g})$), **Concentrated phosphoric acid** (H_3PO_4).

–What Happens: This is an acid-catalyzed addition. The alkene reacts with water (H-OH).

–Product: An alcohol.

Equation Example:



Ethene $\rightleftharpoons$ Ethanol (reversible reaction)

Industrial Use: The primary industrial method for manufacturing ethanol (solvent, fuel).

Addition of Hydrogen Halides (HX)

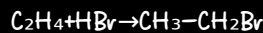
–Reagents: Gaseous hydrogen halides (HCl, HBr, HI).

–Mechanism: Electrophilic addition.

–What Happens: The polar H-X bond adds across the double bond: H^+ (electrophile) adds first, then X^- (nucleophile).

–Product: A mono-substituted halogenoalkane.

Equation Example:



Ethene $\rightarrow$ Bromoethane

Oxidation with Acidified Aqueous Potassium Manganate(VII)

–Reagents & Conditions: **acidified potassium manganate(VII)** solution ($\text{KMnO}_4 + \text{H}_2\text{SO}_4$).

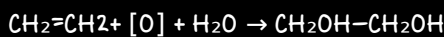
–Mechanism Type: **Oxidation & addition**

–What Happens: The purple MnO_4^- ion oxidizes the C=C bond. The π bond breaks and **OH groups** add to each carbon.

–Observations: **Purple** color disappears (decolorizes).

–Product: A diol (a molecule with two -OH groups on adjacent carbons).

–Equation Example (Ethene):



Ethene $\rightarrow$ Ethane-1,2-diol

Addition of Hydrogenalkanes Major and minor products

Important Note – for unsymmetrical alkenes (e.g. prop-2-ene) two alkanes products are formed (major and minor)

Markovnikov's Rule: Hydrogen adds to the carbon with more hydrogens

Why? : adding H^+ to the carbon with more H's creates the more stable carbocation on the carbon with fewer H's.

–This **stable carbocation intermediate** then reacts with X^- to give the major product.

Therefore: The major product is the one where the halogen (X) is on the more substituted carbon as it is more stable. (tertiary > secondary > primary)

Primary Carbon (1°): A carbon atom bonded to only one other carbon

Example: $\text{CH}_3-\text{CH}_2-\text{CH}_3$ (the first C is bonded to one carbons).

Secondary Carbon (2°): A carbon atom bonded to two other carbon

Example: $\text{CH}_3-\text{CH}_2-\text{CH}_3$ (the second C is bonded to two carbons).

Tertiary Carbon (3°): A carbon atom bonded to three other carbon

Example: $(\text{CH}_3)_3\text{C}-\text{H}$ (The central C is tertiary).

Explaining/Justifying Major and Minor Products:

Identify the intermediate carbocations.

Example: "Product A forms from a tertiary carbocation. Product B forms from a primary carbocation." Compare their stability.

Example: "A tertiary carbocation is more stable than a primary carbocation." State the conclusion.

Example: "Therefore, A is the major product and B is the minor product."

Polymerisation

Mechanism of addition

(halogens and hydrogen halides addition)

Step 1: Electrophilic Attack & Carbocation intermediate Formation:

The alkene's electron-rich π bond attacks the δ^+ atom in the polar molecule (H in H-Br or Br in Br-Br).

The H-Br or Br-Br bond breaks **heterolytically** (unevenly, both electrons go to one atom). H^+ (Br^+) adds to one carbon, forming a new C-H bond.

A **carbocation** forms on the other carbon and a bromide ion (Br^-).

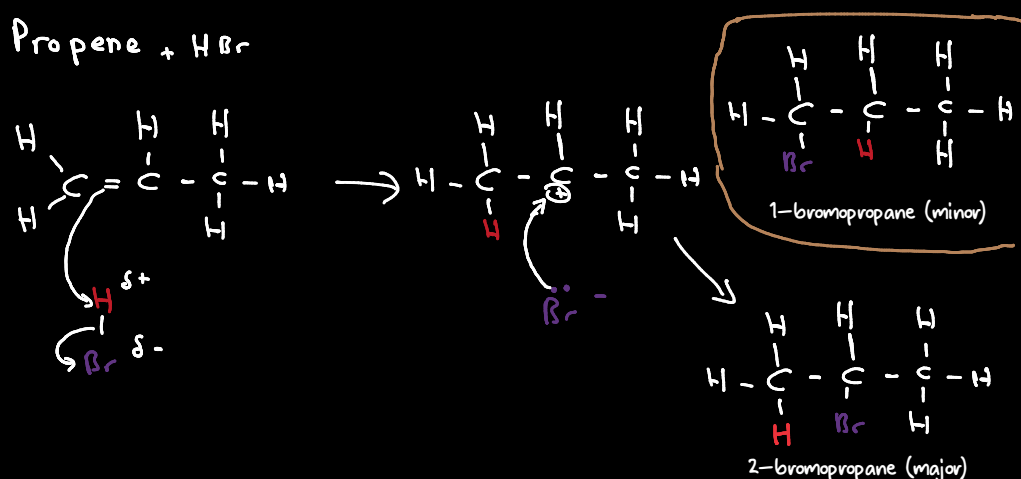
Step 2: Nucleophilic Attack by the Halide Ion

The carbocation (electron-deficient) is attacked by the nucleophilic Br^- .

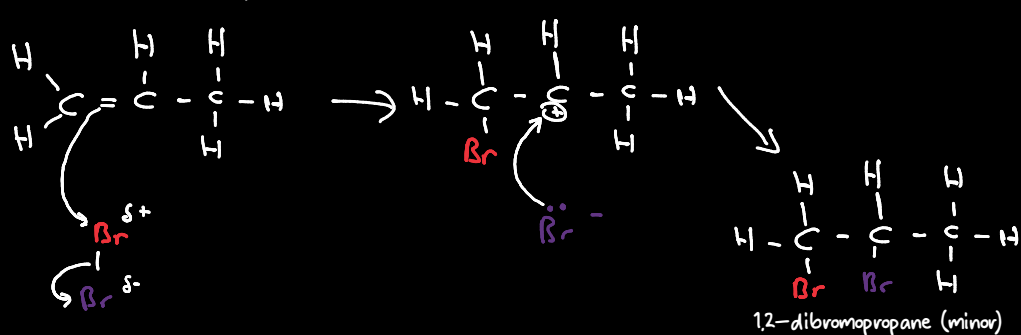
A new C-Br bond forms, yielding the final haloalkane.

note: in case hydrogen halides react with **asymmetrical** alkenes two products are formed (rule mentioned in the previous page)

Propene + HBr



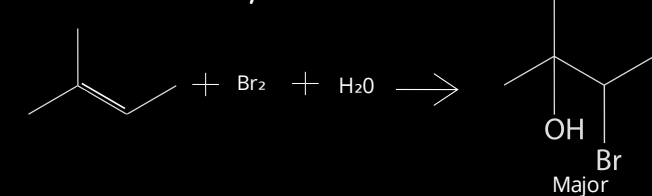
Propene + Br₂



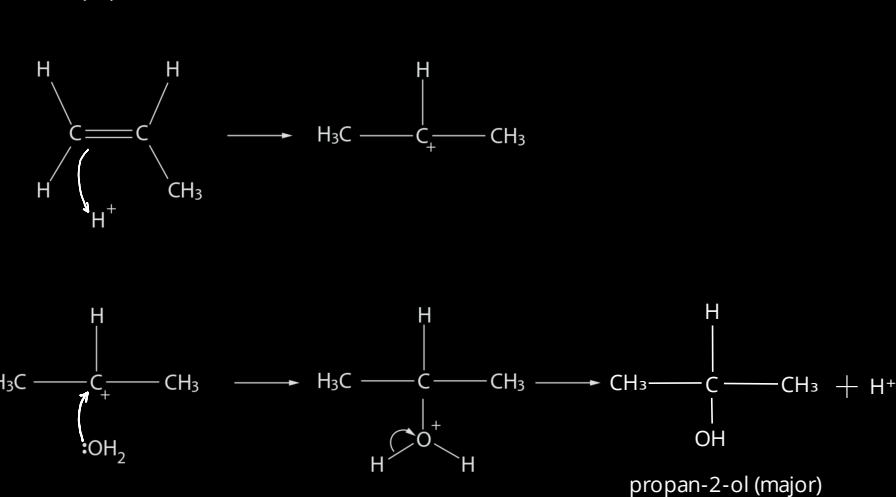
The double bond (π -bond) of the alkene repel electrons (distorts the electron cloud) in the bromine molecule / induces a dipole making the closer atoms slightly positive

Honorable mentions

Bromine water + 2-methylbut-2-ene



Steam + propene



Introduction

Definition: Polymerisation is a chemical reaction where many small molecules (monomers) join together to form a very large molecule (polymer).

For alkenes, this is specifically **addition polymerisation**.

—Reagents/Conditions: High pressure & temperature, often with a catalyst.

—Mechanism Type: Chain-growth addition polymerisation.

—What happens: The C=C double bond opens, and monomers link together by forming new single bonds.

—Key Feature: The polymer retains all the atoms from the monomer—no other product is formed.

The Naming System

Naming a polymer is straightforward once you know the monomer.

Rule: "Poly(monomer name)"

Simply put the name of the monomer in brackets after the prefix "poly-".

Addition vs Condensation

Addition Polymerisation:

—Monomers add together **without losing** any atoms. It's like linking train carriages together.

—Needs **unsaturated monomers**.

—Only **one product** (higher atom economy)

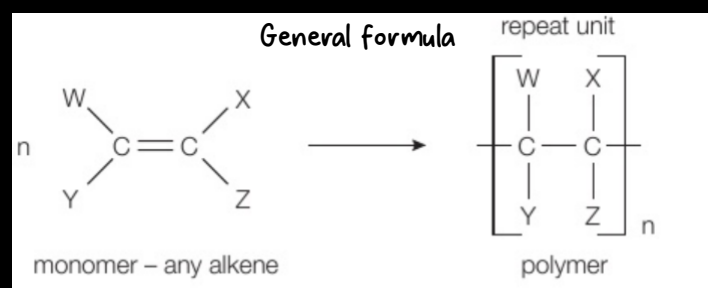
Condensation Polymerisation:

—Monomers join by releasing a small molecule (usually H_2O , HCl , or NH_3). **Some** material (water) is **lost** in the process.

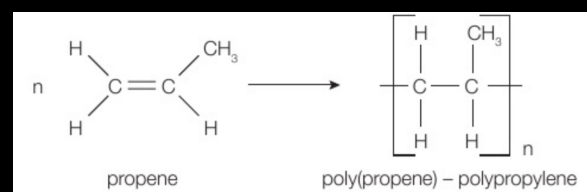
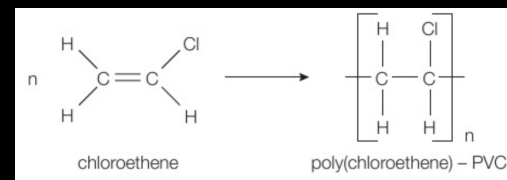
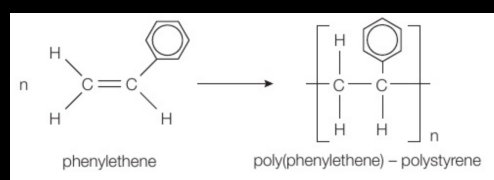
—Needs **two different functional groups** on the monomers (e.g. $-COOH$ & $-OH$, or $-NH_2$ & $-COOH$).

—Two products

(Further knowledge of condensation polymerization is not required)



Examples



Monomer vs polymers

- Monomers are simple molecules that have weak intermolecular forces
- Polymers are macromolecules that have strong intermolecular forces

The Problem with Polymers

The widespread use of synthetic polymers (especially non-biodegradable plastics) has led to a major waste management crisis. These materials are **3 durable, 4 cheap, 5 versatile, light and unreactive**, but these same properties make them environmentally persistent when discarded.

Key Problems:

Persistence in Landfills: Most polymers are non-biodegradable and can remain in landfills for hundreds of years, taking up valuable space.

Environmental Pollution: Improper disposal leads to litter in terrestrial and marine environments, causing harm to wildlife through ingestion and entanglement.

Toxicity from Incineration: Burning polymers without proper controls releases toxic gases (e.g., sulfur dioxide, hydrogen chloride) and particulates, contributing to air pollution and health issues.

Greenhouse Gas Emissions: Incineration produces carbon dioxide (CO₂), a greenhouse gas, and some polymers release potent greenhouse gases like methane as they slowly break down anaerobically in landfills (decomposition).

Resource Depletion: Most polymers are made from non-renewable fossil fuels (crude oil and natural gas).

Incineration of Polymers

Incineration: A waste treatment process that involves the combustion of organic substances contained in waste materials, converting them into ash, flue gas, and heat.

Advantages of Incineration:

- *Produces energy (heat/electricity) that can be used.
- *Prevents polymers from taking up space in landfills.
- Reduce risk of water/soil pollution

Disadvantages of Incineration:

- *Can produce toxic substances (e.g., sulfur dioxide, dioxins, hydrogen chloride) Particulates/soot may be released, causing respiratory problems.
- *Produces CO₂, a greenhouse gas contributing to climate change.
- Ash residue may contain toxic heavy metals.
- High capital and operating costs for advanced emission-control systems.

Minimise Disposal of Polymers Problems

- *Help develop biodegradable polymers from plant material (polymers that can be re-used / recycled)
- *Remove toxic gases produced by incineration of polymers
- Develop processes to convert polymers back into feedstock (for use in chemical industry)
- Use of IR spectroscopy to separate polymers (for recycling)

Using Renewable Resources

Renewable Resource: A natural resource that can be replenished naturally with the passage of time (e.g., plants, crops) as opposed to finite fossil fuels.

Advantages:

- *Renewable – can be replenished over time.
- Often have a lower carbon footprint than fossil-fuel-based polymers.
- Can support sustainable agriculture/forestry.

Disadvantages:

- *Loss of food resources if food crops (e.g., corn, sugarcane) are diverted for plastic production.
- *Require land to grow, reducing land available for food production (food vs. fuel debate).
- May involve large-scale monoculture, impacting biodiversity.
- Can have high water and fertilizer requirement.

Developing Biodegradable Polymers

Biodegradable Polymer: A polymer that can be broken down by the natural action of microorganisms/decomposers (such as bacteria and fungi) into water, carbon dioxide, methane, and biomass.

Advantages:

- *Take less time to break down compared to most traditional plastics.
- *Reduce waste going to landfill.
- *Do not require incineration.
- Reduce pollution, litter, and harm to wildlife.
- Break down into non-harmful products (e.g., CO₂, H₂O, biomass).
- Often come from a renewable resource/sustainable.

Disadvantages:

- *loss of food resources (in case of using plants)
- Can be more expensive to produce than conventional polymers.
- *Can not be recycled, incinerated etc...
- Material properties (e.g., strength, durability) may be inferior to traditional plastics.

Recycling of Polymere

Definition: The process of collecting, sorting, processing, and converting waste polymer materials into new products or raw materials.

Advantages of Recycling:

- *Conserves Resources
- *Reduces Energy Consumption: recycling uses less energy
- *Reduces Landfill Waste
- Lowers Carbon Footprint: Reduces greenhouse gas emissions (e.g., CO₂)
- Less toxic substances released

Disadvantages / Challenges of Recycling:

- *Energy is still consumed to make the new product
- *Polymers need to be stored (which is expensive)
- Limited Recyclability: not all polymers can be recycled (they need to be separated to be recycled)
- Degradation: Polymers can degrade with each recycling cycle due to chain scission, limiting the number of times they can be effectively recycled.

*: indicates more accurate answer, which are mostly mark scheme answers